Agenda:

→ Curriculum for SQL
→ What is database
→ Why database
→ Types of databases.
→ RDMS
→ Keys (Super Keys)

DSA ⟶ SQL (Non - DSA)

How should you tackle SQL module:

Learn → Attend Lecture → Increase PSP → Contest
↓
Mock Interview

Curriculum

1. Intro to DBMS & SQL - 1
2. Keys
3. Crud → 2
4. Join → 2
5. Aggregates
6. Subqueries
7. Indexes.
8. Transactions → 2

SWE

⑨ Schema Design * → ⑩

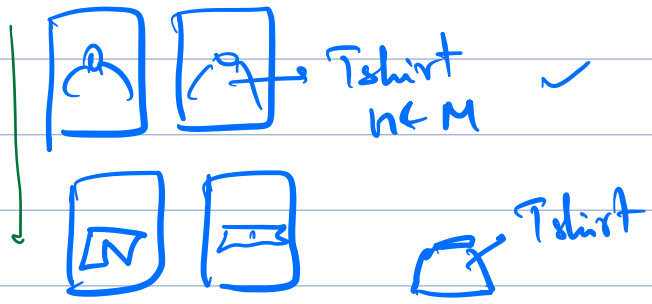
What is not covered

- → Distributed DB's
- → Scalability
- → Sharding
- → Master - Slave
- → Replicatn

} HLD's

Why do we always talk about data?

search order



- → orders details
- → address
- → user details
- → recommdatn → data

→ store data efficiently

→ learners
- → name
- → email
- → pwd
- → age
- → gender
- → psp

} age, gender

csv

Name	psp
sneha	75
vinay	80
Dhem.	85
—	
—	
—	
—	

→ txt, file, csv, excel

→ avg of psp $= \frac{\Sigma \text{ psp}}{\text{Total no of students}}$

Drawbacks (csv, excel, files)

ID	Name	Psp	Attendance (double)	BatchId	password
1	Dipesh	70	80	1	dipesh12
2	David	75	80	1	
3	Dipesh	80	Hello	2	
4	Dhanay	85	90	2	

1. Inefficient
2. Data Integrity
3. Concurrency Issues
4. Security Issues

What is database

→ collection of related data.

Scale

students.csv
batch.csv
instructor.csv
⋮

Army Base Air Base Naval Base

DBMS

↳ Database Management System

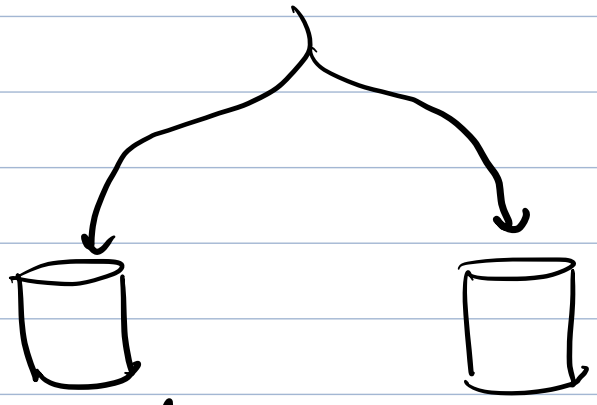
↳ software that allows us to manage a database.

- → Create
- → Read
- → Update
- → Delete

→ DBMS allows us to achieve this efficiently

- → Data Integrity
- → Security
- → Concurrency

Types of Database



Relational Non - Relational

students

id	name	psp	batch-id
1	—	—	1 (SQL)
2	—	—	1 (SQL)

Batch

id	name of b.
1	SQL
2	

500

2
1 (SQL) → ~~Rela~~ DBMS
2
2

10

Relational → store data in tabular format → Rows
→ Columns

→ Non Relational
- → Tables ✗
- → Document ✓
- → Key value ✓
- → graphs ✓

✗ HLD ✓

Properties of RDBMS

Movies	
id	movie name
1	

✗ List<Actor>

Actors	
id	actor Name
1	
2	

movie_actors.

movie_id	actor_id
1	1
1	2

① Represent in collect^n of tables.

ID	Name	Psp	Attendan. (double)	BatchId	password
1	Dipesh	70	80	1	dipesh12
2	Dheeraj	75	80	1	
3	Dipesh	80	Hello	2	
4	Dheeraj	75	80	1	

② Every row should be unique

③ A column should have all values of same data type

ID	Name	Psp	Attendan. (double)	BatchId	password
1	Dipesh	70	80	1	dipesh12
2	Daniel	75	80	1	
3	Dipesh	80	Hello	2	
4	Dheeraj	85	90	2	
5					

throw an error

④ All values should be atomic

Students

id	name	age	phone_number
1	Rushikesh	20	821053, 467899
2	—	—	678992

(5) Ordering of rows is not guaranteed

(6) Ordering cols is " "

(7)

id	name	phone	phone
1	xyz	—	—

Name of every column should be unique.

Keys.

Students.

name	psp	b_id
Suman	80	2
Kanhan	82	2
Keisha	78	1
Rohit	68	2
Suman.	85	2

Update psp of suman to 100

→ uniquely identify a row

Types of keys.

1. Super keys
2. Candidate "
3. Primary "
4. Composite "
5. Foreign "

Students.

unique

id	name	psp	b_id
1	Suman	80	2
2	Kanhan	82	2
3	Krishna	78	1
4	Rohit	68	2
5	Suman.	85	2

b_id	nam
2	SQL

colum	super key
name	✗
b_id	✗
psp	✗
id	✓
id, name	✓
id, psp	✓
email	✓
email, name	✓
phone	✓
phone, name	✓

Super keys: column / combinatn of columns to uniquely identify a row

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Assignments → 15 min MCQ ✓